

Answer A

27) $R = \rho l / A$

The square sheet and the rectangle sheet are connected in series. The square sheet has resistance of $4.0 \, \Omega$, $R = \frac{\rho x}{xt}$ while the rectangle sheet has resistance of

$$\frac{\rho(2x)}{\frac{x}{2}t} = 4(4.0)$$

Total resistance = $20.0 \, \Omega$

Answer D